

Week 1: Python basics

Python syntax and code patterns. Keep it next to you while you work.

Running cells

A variable exists only once its cell has run. **Run All** runs them in order from the top, **Restart** discards every variable, **Clear All Outputs** keeps your code and removes the results. When a notebook misbehaves: Restart, then Run All.

Variables and types

```
n_trials = 240          # int
rt = 0.42               # float
cohort = "CSHL-WT"      # str
correct = True          # bool
missing = None          # NoneType

type(rt)                # <class 'float'>
int("7"), float("0.5"), str(7) # convert
```

Arithmetic

```
+ - * /                / always gives a float
//                    integer division: 7 // 2 is 3
%                     remainder: 7 % 2 is 1
**                    power: 2 ** 3 is 8
x += 1               shorthand for x = x + 1
```

print and f-strings

```
print("value =", rt, type(rt))    # commas add spaces
print(n, end=" ")                 # no new line

f"mean RT {rt:.3f} s"             # 0.420
f"accuracy {acc:.1%}"             # 82.0%
f"{n_trials:,.0f} trials"         # 694,182
f"contrast {c:+.4f}"              # +0.2500
f"{lab:<15} {group:<4} {n}"        # pad left, columns line up
f"{n:>5}"                         # pad right
f"p = {p:.4g}"                   # 4 significant digits
```

Text

```
name = "CSHL" + "_051"
len(name)                        # 8 characters
name.upper(), name.lower()
name.split("_")                  # ['CSHL', '051']
"CSHL" in name                   # True
```

Lists

```
rts = [0.31, 0.42, 0.28]
rts[0], rts[-1]                 # first, last
rts[0:2]                        # from 0, up to but NOT 2
rts.append(0.55)                # add at the end
rts.sort(); rts.reverse()       # in place, returns nothing
last = rts.pop()                # remove and hand back
len(rts), sum(rts), max(rts), min(rts)
sorted(rts)                     # a sorted copy
0.42 in rts                     # True
```

Dictionaries

```
mouse = {"id": "CSHL051", "lab": "CSHL"}
mouse["id"]                     # look up by key
mouse["age"] = 14               # add or overwrite
"lab" in mouse                  # True
mouse.keys(), mouse.values()

for key, value in mouse.items():
    print(key, value)
```

Tuples

```
point = (0.25, 0.83)
x, y = point                    # unpacking
key = (lab, group)              # usable as a dict key
```

A tuple cannot be changed after it is built, which is why it can be a key and a list cannot.

Conditionals

```
if signed_contrast > 0:
    side = "right"
elif signed_contrast < 0:
    side = "left"
else:
    side = "zero contrast"

== !=                equal, not equal
< > <= >=          comparisons
and or not          combine conditions
is True, is None    identity, used for True and None
```

Loops

```
for rt in rts:                # each item
    print(rt)

for i in range(5):             # 0 1 2 3 4
for n in range(0, 21, 2):      # start, stop, step
```

```
for _ in range(100):           # repeat, index unused

for i, rt in enumerate(rts):    # index and item
for lab, grp in zip(labs, groups): # two lists at once

while trials_left > 0:
    trials_left -= 1

break        # leave the loop now
continue    # skip to the next item
```

Pattern: count into a dictionary

```
counts = {}
for trial in trials:
    mouse = trial["mouse"]
    if mouse not in counts:
        counts[mouse] = 0
    counts[mouse] += 1

for mouse, n in counts.items():
    print(f"{mouse}: {n} trials")
```

Pattern: average per group

```
total = {}
n = {}
for i in range(len(labs)):
    lab = labs[i]
    if lab not in total:
        total[lab] = 0
        n[lab] = 0
    total[lab] += ages[i]
    n[lab] += 1

for lab in total:
    print(f"{lab:<12} {total[lab] / n[lab]:.1f} weeks")
```

Pattern: count with conditions

```
n_correct = 0
n_wrong = 0
for trial in trials:
    if trial["mouse"] != "CSHL046":
        continue # skip the rest of this turn
    if trial["correct"] is True:
        n_correct += 1
    else:
        n_wrong += 1

accuracy = n_correct / (n_correct + n_wrong)
```

Reading an error

Read the **last** line first: it names the problem.

NameError	a name never defined, often a typo
TypeError	an operation between incompatible types
IndexError	a position that does not exist
KeyError	a key or column that is not there
SyntaxError	Python cannot read the line at all
IndentationError	a block indented inconsistently

Imports and help

```
help(round)            # the full documentation
round?                 # a quick summary, in Jupyter

import math, random
import numpy as np
from pathlib import Path
from src.qmn_utils import sigmoid

math.sqrt(50), math.log(1000), math.pi
random.seed(1); random.choice(rts)
Path.cwd(), Path("data") / "file.csv"
```

Files

```
with open("summary.txt", "w") as f:
    for key, value in summary.items():
        f.write(f"{key}: {value}\n")

with open("summary.txt", "r") as f:
    text = f.read()
```

"w" overwrites, "a" appends, "r" reads. The file closes itself.

A first look at pandas

```
mice = pd.read_csv(path)
mice.head(), mice.shape, mice.info()
mice["lab"]                # one column
mice["lab"].tolist()       # as a plain list
mice["age"].mean()         # by position
mice.iloc[0]               # by position
mice.loc[0, "lab"]         # by label
```